LEVELHULL · USER REQUIREMENTS

LevelHull user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade), drawn from the users, stakeholders and constraints of LVH-PRB-001

LevelHull user requirements

What fishers, owners, builders and safety programmes need, written from their side. Product requirements that meet these needs are in LVH-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix (`03b-traceability.md`). All are proposals for review with the first co-design partner. Method: STANDARDS section 22.

> **Safety:** LevelHull is flotation for the boat, never a substitute for life jackets. UR-015 makes that a requirement in its own right.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Stays afloat when swamped	When a wave fills our canoe, it stays afloat with us holding on, the motor, the nets and the catch, long enough to be bailed or rescued	High	Canoe fishers and crews (PRB Table 1a)
UR-002	Performance	Gunwale clear of the water	The swamped canoe floats with its gunwale out of the water so we can bail it rather than watch the waves refill it	High	Canoe fishers and crews
UR-003	Safety	Floats upright and level	The swamped canoe floats upright and level and does not roll over on us	High	Canoe fishers and crews
UR-004	Usability	Can be bailed out	Two of us can bail it out with what is aboard and get going again	High	Canoe fishers and crews
UR-005	Usability	Room to work	It leaves room to work, stow the catch and handle the nets	High	Canoe owners
UR-006	Environment and durability	Keeps its lift	It keeps its lift in water, strong sun, fuel and fish slime	High	Canoe owners
UR-007	Safety	Stays fixed	It stays fixed to the canoe when the canoe is full of water, struck by waves or on its side	High	Canoe fishers and crews
UR-008	Deployment and assembly	Fitted locally in a day	A local builder can fit it in a working day with hand tools, without changing the hull shape and without new holes below the waterline	High	Local boatbuilders
UR-009	Environment and durability	Lasts	It lasts at least a year of normal use without cracking, tearing or losing lift	Medium	Canoe owners
UR-010	Cost	Affordable	The kit costs what an owner or a safety programme can pay	Medium	Canoe owners; fisheries departments and NGOs
UR-011	Performance	Reserve	It still copes with an extra person holding on or a wet net more than planned	Medium	Canoe fishers and crews
UR-012	Safety	Damage tolerant	If one piece is torn or lost the canoe still floats	Medium	Fisheries departments, BMUs and NGOs
UR-013	Safety	Crew on one side	It stays upright when all of us grab the same side	High	Canoe fishers and crews
UR-014	Usability	Out of the way	It does not hinder beaching, landing alongside other canoes, paddling, motoring or hauling nets over the side	Medium	Canoe owners; canoe fishers and crews
UR-015	Regulatory	With life jackets, never instead	The kit and everything said about it make clear it is not a life jacket and never replaces one	High	Fisheries departments, BMUs and NGOs

ID	Category	Title	Need	Priority	Source
UR-016	Logistics	Sizes for other canoes	Builders can size it for the canoes they have, of other lengths, beams and motors	Medium	Local boatbuilders; fisheries research institutes

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.